Cirrus Protocol Standardization Guide

Version 2.0. Source: "Protocol Standardization 2.0" workbook, hosted by the DAMP Lab. Drafted and edited by Luciana Lozano and Raven Coffer, Cumulab, Boston University 2026 iGEM Team.

1. Why this guide exists

This guide sets one shared format for writing lab protocols so they can be run the same way in any lab. Within one lab, a protocol is usually written by one member and carried out by peers. Each lab has its own habits, so details that seem obvious to that group are often left out. That causes little trouble inside the lab. When another lab adopts the protocol, the gaps leave doubt about what an action means, which equipment is meant, and which small steps were never written down. The problem is bigger in a cloud lab, where the scientist who designs the experiment is not the technician who runs it, and undocumented details lead to errors and inconsistent results.

To close that gap, the guide standardizes three things: **the structure** of a protocol, **the names** of its materials, and **its language**. Once a technician knows this format, they can read and run any protocol written in it with little to no miscommunication.

2. How to use this guide

Part	What it defines	How to use it
Part A. Protocol structure	The skeleton of the protocol: four fixed Steps (Overview, Preparation, Procedure, Clean Up) and the Sub-Steps that can appear under each.	Keep the Steps in this exact order. Not every protocol needs every Sub-Step, so include only the ones the protocol actually needs. Sub-Steps may be reordered within their Step but must stay under the Step they belong to.
Part B. Materials nomenclature	The dictionary for equipment and consumables.	Wherever an item is mentioned, use its Standardized Name instead of a brand name or informal term. The "Also known as" column lists the alternates each standard name replaces.
Part C. Specific tasks	The dictionary for lab actions. Each action has an attention type, a definition and a fill-in-the-blank Sentence Structure.	Copy the Sentence Structure into the protocol and replace each [bracketed] field with the specifics. The Example column shows how the finished sentence should read.

Attention types tell the technician how much oversight a step needs:

- **Active**: hands-on the whole time.
- **Walk-away**: start it and leave.
- **Monitor**: check on it periodically.

A task marked "walk-away or monitor" can be either, depending on the protocol.

For a reference of what a standardized protocol looks like, see the standardized protocols published by the DAMP Lab.

Part A. Protocol structure

Step: Overview

Sub-Step	Definition	Example
Title	The protocol's name, used to identify it in the lab notebook, file system, and any publication registries (ex. protocols.io).	—
Abstract	This is short summary of what the protocol does (1-2 sentences) followed by a concise description of the expected outcome (1-2 sentences). It gives the reader an initial gist without reading the full protocol.	Example abstract from DAMP Lab's Wizard® MagneSil® Plasmid Purification Kit Protocol: The Wizard® MagneSil® Plasmid Purification System by Promega enables rapid, high-throughput isolation of plasmid DNA in a multiwell format using paramagnetic particles for both lysate clearing and DNA capture. The resulting plasmid DNA is of sufficient quality for downstream applications such as DNA sequencing and restriction enzyme digestion.
Materials	A complete list of everything needed to run the protocol. Split into two categories organized in bulleted items: consumables and equipment. Consumable section has a kit-specific header as well, listing all the reagents from the kit. Equipment section should consist of general equipment/device names. Include a table listing lab-specific equipment brand names and models that were used to validate the protocol. Lab-specific equipment table title: "Table 1. The specific devices used by the XXXX Lab for the XXXX protocol."	—
References	Citations to source protocols, kits, product numbers, or external documentation the protocol is based on or depends on.	—
Acknowledgements	Credit for everyone involved in creating and maintaining the protocol (drafting, editing, uploading).	XXXX and XXXX initially drafted the protocols based on XXXX and an internal protocol developed by XXXX. XXXX edited the protocol. XXXX uploaded the protocol to protocols.io.
Warnings	Safety-critical cautions the user must know before starting.	—
Guidelines	General best-practice tips that apply throughout the entire protocol, not tied to any single step.	Perform all work close to the bunsen burner.

Sub-Step	Definition	Example
Before Start	Prerequisite conditions or information the lab technician must confirm before beginning the protocol.	Know growth conditions of sample/sample concentration.

Step: Preparation

Sub-Step	Definition	Example
Overview	A graphical summary of the preparation workflow, to orient the user before starting individual steps. This is essentially a graphical representation of the abstract.	—
Setup	General environment and reagent preparation.	Bring reagents to room temperature.
Label	Marking tubes, plates, or containers with identifying information (sample name, initials, date) before use.	—
Initialize equipment	Powering on, logging into, or starting the programs/instruments that will be used later in the protocol.	—
Decontamination	Cleaning the work surface and tools with the appropriate decontaminant(s) before starting hands-on work.	Wipe benches and pipettes with DNA Away, RNase Away. Afterwards wipe benches and pipettes with 70% ethanol.

Step: Procedure

Sub-Step	Definition	Example
Sample Preparation	Includes actions performed directly on the sample (ex. adding reagents, transferring, mixing, resuspending) to carry out the core protocol.	—
Run Configuration	Selecting or entering the settings/parameters an instrument or software will use for the run.	Running configuration on opentrons, blanking NanoDrop, etc.
Sample Submission	Packaging and delivering a prepared sample to its next destination.	—
Incubation	A defined waiting period where the sample is held at a specific temperature and/or agitation for the reaction or growth to occur.	—
Capture Data	Recording a measurement, reading, or output value generated by the procedure and saving it for later reference.	Saving data from the NanoDrop.
Inactivation	Inactivating a protein/enzyme with heat, pcr cleanup, loading dye, etc.	—

Step: Clean Up

Sub-Step	Definition	Example
Disposal	Discarding used consumables, samples, or waste into the correct, clearly labeled waste stream.	—
Decontamination	Cleaning the work surface and tools with the appropriate decontaminant(s) after finishing hands-on work.	Wipe benches and pipettes with DNA Away, RNase Away. Afterwards wipe benches and pipettes with 70% ethanol.
Storage	Returning samples or reagents to their designated storage condition (ex. temperature, container) for future use.	—
Equipment Shutdown	Turning off, extinguishing, or otherwise safely powering down any equipment used during the protocol once work is complete.	Extinguishing the bunsen burner, closing software, powering off instruments, etc.

Part B. Materials: standardized nomenclature

Use the **Standardized Name** in every protocol. If your protocol uses a name from the "Also known as" column, replace it with the standard name.

Consumable

Subcategory	Standardized name	Description	Also known as
Pipette tips	Opentrons Pipette Tips	Disposable pipette tips sized for the Opentrons automated pipette arms.	—
Pipette tips	Filtered Pipette Tips	Disposable filtered pipette tips used with manual pipettes to prevent aerosol contamination.	—
Tubes & columns	15 mL Snap-Cap Tubes	Sterile 15 mL tubes used to grow individual overnight bacterial cultures.	15 mL snap-cap culture tubes
Tubes & columns	Microcentrifuge Tubes	Snap-cap 1.5 mL tubes used to hold standards, cell pellets, or other reactants.	1.5 mL snap-cap tubes; 1.5 mL Eppendorf tubes
Tubes & columns	Microcentrifuge Strip-Tubes	Connected strip of microcentrifuge tubes (typically ranging 0.5-2 mL) used to hold multiple diluted samples in a fixed order for submission.	—
Tubes & columns	50 mL Conical Tube	Large conical tube that are sometimes used as an outer, padded shipping container for smaller sample tubes.	—
Tubes & columns	Spin Column	Silica spin-column and matched collection tube that isolates and purifies DNA, RNA, or proteins with the combined use of a centrifuge.	Collection tube

Subcategory	Standardized name	Description	Also known as
Plates	Clear Round-Bottom 96-Well Plate	Clear, round-bottom 96-well plate commonly used for OD600 growth-density reads; distinct from the black optical-bottom plate and the deep-well plate.	—
Plates	96-Well Black Optical-Bottom Plate	Black-walled, optical-bottom 96-well plate.	NUNC 96 wellplate optical bottom black
Plates	96-Well Deep-Well Plate (2 mL)	Deep-well 96-well plate used to hold larger liquid volumes.	Deep-well (2ml) 96-well plate; 96 deep well plate; 96-well deep-well plate (2 mL)
Liquids & reagents	80% Ethanol	Diluted ethanol.	—
Liquids & reagents	100% Isopropanol	Undiluted isopropanol.	—
Liquids & reagents	Nuclease-Free Water (NFW)	Water certified free of nucleases, used to dilute or elute nucleic acid samples without risk of degradation.	Nuclease Free Water; NFW
Liquids & reagents	DI Water	Deionized water used for general cleaning purposes (ex. wiping instrument pedestals) where nuclease-free grade is not required.	Deionized water
Liquids & reagents	Antibiotic	Selective antibiotic (ex. ampicillin, kanamycin) added to culture media, matched to the plasmid's resistance marker.	Appropriate antibiotic (ampicillin, kanamycin, etc.)
Liquids & reagents	Liquid Culture Media	Bulk liquid growth medium (ex. LB, M9) used to culture bacteria; combined with antibiotic before aliquoting into culture tubes.	Appropriate liquid culture media (LB, M9, etc.)
Culture & inoculation supplies	Sterile Inoculation Loops	Single-use sterile loops used to transfer bacteria from a colony or glycerol stock into culture media.	Sterile loops; inoculation loop
Cleaning	KimWipes	Lint-free absorbent wipes used to blot, dry, or clean surfaces and instrument components.	—
Waste containers	Bleach Waste Beaker	Dedicated, labeled beaker used to collect 10% bleach-treated biological waste (ex. discarded supernatant, spent cultures) separate from other waste streams.	—
Waste containers	Ethanol Waste Beaker	Dedicated, labeled beaker used to collect spent ethanol/alcohol-containing wash waste separate from biohazard waste.	—
Packaging & shipping	Small Shipping Bag	Sealable bag used to package the sample container and order form together for vendor submission.	Small bag
Packaging & shipping	Heat Seal	Heat-applied film used to seal a 96-well plate for storage or shipment.	Plate sealer
Temperature Control	Dry Ice	Solid CO_2 used to keep glycerol stocks frozen and minimize thaw exposure during handling.	—

Equipment

Subcategory	Standardized name	Description	Also known as
Liquid handling	Manual Pipettes	Hand-held pipettes used for manual liquid transfers not performed by the automated robots.	—
Liquid handling	Serological Pipette	Graduated glass or plastic pipette used with a pipette pump/controller to accurately dispense larger volumes (mL range).	Stripette
Centrifuges	Tabletop centrifuge	Tabletop centrifuge commonly fitted with 96-well plate adapters, used to pellet cells and clarify lysate in plate format (rated 1,500 × g).	Benchtop centrifuge
Centrifuges	Minicentrifuge	Very small, low-spin centrifuge unit for quick spin-downs. Commonly used to collect droplets to the bottom of a tube	—
Centrifuges	Microcentrifuge	Standard tube-format benchtop microcentrifuge capable of high-speed spins.	—
Mixing	Vortexer	Benchtop device that rapidly agitates a tube in a circular motion to mix contents or resuspend a pellet.	—
Mixing	Benchtop Microplate Shaker	Benchtop shaker fitted with plate holders, used to agitate/resuspend or dry a 96-well plate at controlled RPM and temperature.	Microplate shaker fitted with plate holders; shaker
Magnets	96-Well Magnet	Magnetic stand that holds paramagnetic beads in place against the well wall of a 96-well plate, allowing supernatant to be removed without disturbing bound beads.	Magnetic Workstation
Incubators	Shaking Incubator	Incubator with an orbital or linear shaking platform used to grow liquid cultures at controlled temperature with continuous agitation for aeration.	—
Incubators	Static Incubator	Incubator without agitation, used to grow samples at a controlled temperature.	—
Analytical	Microplate Reader	Benchtop instrument that reads absorbance (or other optical signal) across all wells of a microplate at a specified wavelength; used for OD600 and other plate-based reads.	plate reader
Analytical	NanoDrop	Benchtop micro-volume, 8-pedestal spectrophotometer used to measure nucleic acid or protein concentration and purity ratios (A260/A280, A260/A230), controlled via the NanoEight software.	NanoDrop Spectrophotometer
Analytical	Qubit Fluorometer	Benchtop fluorometer used for fluorescence-based DNA/RNA/protein quantification.	Qubit
Bunsen Burner	Bunsen Burner	Gas burner used to create a sterile convection field for aseptic technique; work is performed close to the flame during sample handling.	—
Glassware	250 mL Erlenmeyer Flask	Sterile glass flask commonly used to prepare bulk culture media plus antibiotic before aliquoting into individual culture tubes.	—

Opentrons

Subcategory	Standardized name	Description	Also known as
Robot	Opentrons Flex	Automated liquid-handling robot deck that executes pipetting, mixing, and plate-transfer steps per a programmed deck layout.	—
Modules	Opentrons Heater-Shaker Module	Deck-mounted module that heats and/or shakes a plate under robotic control, fitted with a universal flat plate attachment.	—
Pipettes	Opentrons 8-Channel 1000 µL Pipette	Automated multi-channel pipette arm mounted on the Opentrons Flex, used for parallel transfers across plate columns.	Opentrons 1000 uL 8-channel pipette arm
Pipettes	Opentrons 1-Channel 50 µL Pipette	Automated single-channel pipette arm mounted on the Opentrons Flex, used for precise low-volume transfers.	Opentrons 50 uL 1-channel pipette arm
Labware	Opentrons pipette tip rack	Tip rack for the opentrons of multiple volumes (50 µL, 200 µL, etc)	Opentrons tiprack
Labware	Opentrons 24-Tube Rack	Deck-mounted rack holding 24 standard tubes for robotic pipetting access.	Opentrons 24 tuberack holder
Labware	Opentrons 12-Well Reservoir (22 mL)	Deck-mounted multi-channel reservoir used to hold bulk reagent for robotic pipetting.	—

Part C. Specific tasks: standardized action vocabulary

Fields in [brackets] are replaced with the specifics from the protocol. Where a task has more than one sentence structure, pick the one that matches the intent of the step.

Cleaning

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
wipe	—	active	Cleaning the work surface or material with a decontaminant (ex. 70% ethanol, DNA Away) before/after use.	Wipe [surface] with [decontaminant].	2 minutes	Wipe benches and pipettes with DNA Away, RNase Away. Afterwards wipe benches and pipettes with 70% ethanol.
discard	—	active	Disposing of used consumables or waste into the appropriate labeled waste stream.	Discard [item] into the [waste stream].	—	Discard the clearing plate into the biohazard waste.
empty	—	active	Pouring out liquid waste into a drain or designated collection container.	Empty [liquid] into the [collection container].	—	Empty guanidine/alcohol waste into the ethanol waste collection site.

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
empty	—	active	Pouring out liquid waste into a drain or designated collection container.	Empty [liquid] down the drain.	—	Empty the bleached cultures and the bleach waste beaker down the drain.
store	—	active	Placing a sample or reagent into designated storage conditions until further use.	Store [sample] at [temperature].	—	Store the collection plate at -20°C.
rinse	—	active	Washing a surface or item briefly with a solvent (ex. water, ethanol) to remove residue.	Rinse [item] with [solvent].	—	Rinse the beaker with ethanol.
package	—	active	Preparing a sample or item for shipment by securing it appropriately (ex. padding, sealing). This action specifies how to package the item (ex. wrap in parafilm, put in iced box, etc).	Package [item] for [purpose] by [action] per [vendor] guidelines.	—	Package the DNA extracts for sequencing by placing them in insulated mailers per [Vendor name] guidelines.
package	—	active	Preparing a sample or item for shipment by securing it appropriately (ex. padding, sealing). This action specifies how to package the item (ex. wrap in parafilm, put in iced box, etc).	Package [item] for [purpose] by [action].	—	Package the DNA extracts for sequencing by placing them in insulated mailers.
submit sample to an external vendor	—	active	Placing a prepared sample in the designated dropoff/shipping location by the specified deadline.	Submit [sample] to [dropbox/carrier] by [deadline].	—	Submit the sequencing samples to FedEx by 3:00pm Friday 09/04/2026.
submit sample to an external vendor	—	active	Electronically sending a sample. The recipient could be a website, email address, etc. If there is no deadline omit last part.	Send [sample] as [file format] to [recipient] by [deadline]	—	Send the plate map as an XLSX to handle@gmail.com by 3:00pm Friday 09/04/2026.

Equipment Use

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
centrifuge	—	walk-away	A benchtop instrument that spins samples at high speed to pellet cells, debris, or beads via centrifugal force. Specify the units (RCF or RPM).	Centrifuge [sample] at [speed] for [time].	—	Centrifuge the lysis plate at 3500 x g for 5 min.
preset	—	monitor	Presetting a piece of equipment to a specific temperature or setting.	Preset [equipment] to [temperature/other setting]	—	Preset the microplate heater to 35°C

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
bunsen burner	—	active	Igniting a Bunsen burner to create a sterile convection field.	Light the Bunsen burner and work within its sterile field.	30s	Light the Bunsen burner and work within its sterile field.
spin	—	monitor	A brief, typically lower-volume centrifugation step performed in a compact/mini centrifuge, often to collect droplets or briefly pellet material.	Spin [sample] at [speed] for [time].	—	Spin the microcentrifuge tubes at 1,000 x g for 15 sec.
plate shaker	unheated	walk-away or monitor	An instrument that agitates a microplate at a set speed (RPM) to mix or resuspend contents evenly.	Shake [plate] at [RPM] for [time].	—	Shake the lysis plate at 1800 rpm for 3 min.
plate shaker	heated	walk-away or monitor	Using the plate shaker's heated setting.	Dry [plate] on the shaker at [temp], [RPM], for [time].	—	Dry the binding plate on the shaker at 65°C, 300 rpm, for 10 min.
microwave	—	walk-away or monitor	A designated lab microwave used to rapidly heat or melt reagents (ex. agarose).	Microwave [reagent] for [time].	—	Microwave the gel mix for 45 sec.
electrophoresis rig	—	walk-away	An apparatus that applies an electric field across a gel to separate nucleic acids or proteins by size.	Run the electrophoresis rig at [voltage] for [time].	—	Run the electrophoresis rig at 100 V for 1 hr.
gel image	—	walk-away	An imaging system used to visualize and photograph stained gels (ex. under UV or blue light).	Image the gel using [imager settings].	—	Image the gel using the blue light settings.
magnet	place on magnet	active	Positioning a plate on a magnetic stand so paramagnetic beads (with bound DNA/target) pellet against the well wall, allowing surrounding liquid to be removed without disturbing the beads.	Place the [plate name] on the magnet for [time].	—	Place the binding plate on the magnet for 1 min.

Mixing

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
invert	—	active	Turning a closed container upside-down and back up repeatedly to mix its contents without introducing shear force or bubbles.	Mix [thing] by inverting [container] [# of times].	—	Mix the reagent by inverting the bottle 5 times.
pipette to mix	gentle	active	Drawing liquid up and expelling it slowly through a pipette tip to mix without introducing bubbles or shear stress on sensitive samples.	Gently pipette [sample] up and down [# times] to mix.	—	Gently pipette the ligation reaction up and down 5 times to mix.

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
pipette to mix	strong	active	Drawing liquid up and expelling it repeatedly through a pipette tip to fully disperse more viscous reagents. Also written as pipette to mix or mixing by pipetting	Pipette [sample] up and down [# times] to mix.	—	Pipette the resuspended pellet up and down 10 times to mix.
manually shake	—	active	Manually shaking a bottle up and down to mix	Manually shake [bottle]	—	Manually shake the reagent bottle.
swirl	—	active	Gently rotating an open or closed container in a circular motion by hand to mix contents without vigorous agitation.	Swirl [container] to mix.	—	Swirl the beaker to mix.
flick	—	active	Sharply tapping the side or bottom of a tube with a finger. Commonly used to loosen a pellet before resuspension.	Flick [tube] to loosen [thing].	—	Flick the tube to loosen the pellet.
vortex	—	active	A device that rapidly agitates a tube via circular motion to mix contents or resuspend a pellet.	Vortex [reagent/tube] for [time].	—	Vortex the resuspended pellet for 30 sec.
combine	—	active	Mixing two or more substances together in a single container such as tube, well, etc. This broad term means that the way in which the substances are mixed (pipetting, inverting, swirling, etc) is not critical to the protocol.	Combine [substance 1] and [substance 2] in [container].	—	Combine the substrate and catalyst in a reaction tube.

Preparatory

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
labeling	—	active	Marking tubes, plates, or containers with sample identifiers to ensure traceability throughout the protocol.	Label the [container] as [name].	—	Label the standard 96 well plate as Final Collection Plate.
pipette	—	active	Using a pipette to precisely add a measured volume of reagent to a sample or well.	Pipette [volume] of [reagent] to [location].	—	Pipette 100 uL of buffer to each well.
add	precise volume	active	Adding a reagent or solution to a sample with or without using a pipette (ex. pouring, dispensing from a bottle).	Add [volume] of [reagent] to [location].	—	Add 2 mL of media into each well.
add	ratio-based	active	Adding a reagent to a bulk solution at a specified dilution ratio (ex. antibiotic to culture media) rather than a fixed volume.	Add [reagent] to [solution] at a 1:[ratio] ratio.	—	Add DNase to the lysis buffer at a 1:100 ratio.

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
aliquot	—	active	Using a serological or standard pipette to dispense an equal, defined volume of a liquid into each of several containers.	Aliquot [volume] of [liquid] to each [container].	—	Aliquot 5 mL of culture to each conical tube.
transfer	—	active	Moving a defined volume of liquid from one container or plate to another.	Transfer [volume] of [liquid] from [source] to [destination].	2 minutes (usually)	Transfer 500 uL of supernatant from the tube to a fresh microcentrifuge tube.
let it sit/rest/thaw	—	monitor	Passively waiting while a sample or solution remains undisturbed to reach equilibrium, complete a reaction, or thaw.	Let [sample] [sit/rest/thaw] for [time].	—	Let the sample sit for 5 min.
cover	—	active	Sealing or covering a container or plate to prevent contamination, evaporation, or spillage.	Cover [container] with [seal/lid].	—	Cover the plate with an adhesive seal.
cooling	—	monitor	Reducing sample temperature, either passively (bench/refrigerator) or actively (ice bath).	Cool [sample] at [temperature/on ice] for [time].	—	Cool the sample on ice for 10 min.
measure	weigh (mass)	active	Determining the mass of a sample or reagent using a balance.	Weigh [thing] in [units] and record mass.	—	Weigh the dried pellet in milligrams and record mass.
measure	measure (volume, absorbance, etc)	active	Determining a quantitative property of a sample (volume, absorbance, concentration) using the appropriate instrument.	Measure [property] of [sample] using [instrument].	—	Measure the volume of the eluate using a graduated cylinder.
Wiping/Dab	—	active	Using an absorbent wipe (ex. KimWipe) to blot or remove excess liquid from a surface.	Dab [surface] with a [KimWipe/paper towel] to remove excess liquid.	—	Dab the collection tube with a KimWipe to remove excess liquid.
aspirate	—	active	Using a pipette to precisely draw off and remove a specific volume of supernatant or liquid without disturbing other components.	Aspirate [volume] of [liquid] from [current location] and discard into the [waste stream] waste container.	—	Aspirate ~100 µL of spent wash from each well and discard into the guanidine/alcohol waste container.
inoculate	scrape	active	Using a sterile tool to remove a small amount of solid/frozen material (ex. from a glycerol stock to start a new culture).	Scrape [culture] with [tool] and inoculate [media].	—	Scrape the glycerol stock with a sterile pipette tip and inoculate 5 mL of LB broth.
inoculate	spread	active	Distributing a liquid culture evenly across the surface of a solid media plate using a sterile spreader.	Spread [volume] of [culture] evenly on [plate].	—	Spread 50 uL of culture evenly on the agar plate.

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
inoculate	pick colony	active	Using a sterile loop to touch and lift a single, isolated bacterial colony from a solid media plate, avoiding contact with neighboring colonies.	Pick a single colony from [plate] using a sterile loop and inoculate into [culture tube].	—	Pick a single colony from the LB agar plate using a sterile loop and inoculate into the culture tube.
inoculate	glycerol stock scrape	active	Using an inoculation loop to gently scrape the surface of a frozen glycerol stock (kept on dry ice) until a small amount of material is visible on the tip, then transferring it into culture media.	Scrape the glycerol stock with a sterile loop and inoculate into [culture tube].	—	Scrape the glycerol stock with a sterile loop and inoculate into the culture tube.
dilute	—	active	Reducing a sample's concentration to a target value by adding a defined volume of diluent (ex. water or buffer), calculated from the sample's measured concentration.	Dilute [sample] to [target concentration] by adding [volume] of [diluent].	—	Dilute the DNA extract to 50 ng/uL by adding 30 uL of nuclease-free water.
visual QC check	—	monitor	Visually confirming an expected color change, texture, other visual change in a reaction as a quality checkpoint before proceeding.	Confirm [expected visual change] before proceeding.	—	Confirm the gel has fully solidified before proceeding.
evaporate	open-cap air dry	monitor	Leaving a tube or column open for a defined time to allow residual solvent (ex. ethanol) to evaporate.	With the [cap] open, let [solvent] evaporate for [time].	—	With the cap open, let the ethanol evaporate for 5 min.
execute	secondary protocol	active	Used when another protocol needs to be performed as one of the steps within the current protocol	Execute [secondary protocol].	—	Execute the Restriction Digest protocol.
execute	secondary protocol	active	Used when secondary protocol requires more specifications such as on what is is being executed, and with what equipment.	Follow [secondary protocol] to [what you are measuring] using [equipment].	—	Execute the NanoDrop protocol to measure the plasmid yield of each sample using the NanoDrop.

Software

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
Excel	download	active	Saving a copy of data or a file from an online source or instrument to local storage. If the file is attached to the protocol document no need to include "from [source]" as indicated in the sentence structure.	Download [file] from [source].	—	Open the attached ligation calculator spreadsheet.

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
Excel	record	active	Entering measured data into a spreadsheet or log for tracking.	Enter [data] in [file/sheet].	—	Enter the vector and insert lengths (kb) and the measured concentrations (ng/uL) in the yellow cells of the calculator spreadsheet.
Excel	save	active	Storing the current state of a file to prevent data loss.	Save the file as [filename].	—	Save the file as Calculated Ligation Reaction Volumes.
Excel	file	active	Placing a document or dataset into its designated storage location for organization and retrieval.	File [document] in [location].	—	File the sequencing results in the project folder.
NanoDrop	login	active	Signing into the instrument's software using lab credentials.	Log in to [software]	—	Log in to the laboratory computer and launch the NanoEight Software.
NanoDrop	select/click	active	Choosing an option or button within the software interface to proceed.	Select/click [option] in [software].	—	Select dsDNA in the NanoEight Software home page.
NanoDrop	export	active	Saving data from the software in a specified file format for external use.	Export [data] as [file format].	—	Export the plate reader data as a CSV file.
NanoDrop	upload	active	Transferring a file from local storage into the software/instrument system.	Upload [file] to [software].	—	Upload the plate map to the instrument software.
NanoDrop	run	active	Initiating the instrument's measurement/read cycle via the software.	Run [assay/read] in [software].	—	Run the A260/A280 measurement in the NanoDrop software.
NanoDrop	blank	active	Measuring a reference solution (matching the sample's diluent) on the instrument to zero out background absorbance before reading samples.	Add [volume] of [blank] to the center of the selected pedestal and run to blank the instrument.	—	Add 1 uL of nuclease-free water (NFW) to the center of the selected pedestal and run to blank the instrument.
Calibrate	—	active	Adjusting or verifying an instrument's readings against a known reference standard before use.	Calibrate [instrument] using [standard].	—	Calibrate the NanoDrop using nuclease-free water (NFW).
Image	—	active	Capturing a visual record of a sample or plate via the software/instrument's imaging function.	Image [sample] using [instrument].	—	Image the gel using the UV transilluminator.
Close/exit/sign out	—	active	Ending the software session and safely closing the connection to the instrument.	Close/exit [software] and sign out.	—	Close the NanoEight Software and sign out.

Temperature

Task	Sub-task	Attention	Definition	Sentence structure	Time on protocols.io	Example
incubator	shaking incubator	walk-away or monitor	An incubator with a shaking platform that uses controlled temperature and continuous agitation for aeration.	Incubate [sample] in the shaking incubator at [temp] for [time].	—	Incubate the bacterial culture in the shaking incubator at 37 °C for 16 hr.
incubator	static incubator	walk-away or monitor	An incubator without agitation, commonly used to grow samples (ex. on solid media plates) at a controlled temperature.	Incubate [sample] in the static incubator at [temp] for [time].	—	Incubate the agar plates in the static incubator at 37 °C for 16 hr.
incubator	heat shock	active	Briefly exposing cells to an elevated temperature (typically 42°C) for a defined, short duration.	Heat shock [sample] at [temp] for [time].	—	Heat shock the cells at 42 °C for 45 sec.
incubator	ice bath	walk-away or monitor	Placing a sample on ice or in an ice-water slurry to keep it cold and slow any biological activity.	Place [sample] in an ice bath for [time].	—	Place the enzyme mix in an ice bath for 2 min.
hot plate	—	walk-away or monitor	A flat heated surface used to warm containers or reagents from below.	Place [container] on the hot plate at [temp] for [time].	—	Place the beaker on the hot plate at 60 °C for 10 min.
heating block	—	walk-away or monitor	A metal block with wells that heats tubes directly via conduction, used to incubate tubes at a set temperature.	Place [tube] in the heating block at [temp] for [time].	—	Place the reaction tubes in the heat block at 65 °C for 10 min.
bead-bath	—	walk-away or monitor	A temperature-controlled bath using dry beads (rather than water) as the heat-transfer medium, used to incubate tubes at a set temperature.	Incubate [sample] in the bead-bath at [temp] for [time].	—	Incubate the sample tube in the bead bath at 37 °C for 10 min.

Appendix. Writing a standardized protocol

Manual route. Work through the guide part by part. Reorganize the protocol under the Steps and Sub-Steps in Part A, replace each equipment and consumable with its Standardized Name from Part B, and rewrite every action using the approved verb and Sentence Structure from Part C. Finish with a scientific-accuracy review.

AI-assisted route. Give an AI assistant this guide and the protocol to standardize, then review the result for scientific accuracy before use. The rules the assistant should follow:

1. Organize the protocol under the Steps in the exact order given (Overview, Preparation, Procedure, Clean Up).
2. Put each piece of content under the correct Step and Sub-Step. A Sub-Step may be reordered within its Step but never moved to a different Step.

3. Include only the Sub-Steps the protocol needs, and do not invent Sub-Steps that are not in the source protocol.
4. Replace every equipment and consumable mention with its exact Standardized Name, using "Also known as" to map brand or informal terms.
5. Rewrite every action using the approved verb and exact Sentence Structure, filling the [bracketed] fields with specifics from the source.
6. Never change scientific content: volumes, concentrations, times, temperatures, speeds, reagents, sample identities or the order of operations. Change wording and formatting only.
7. If an action, material or step has no matching entry, do not invent one. Keep the original wording, mark it "NOT IN GUIDE", and list it under "Flags for Review" at the end.
8. Output the fully standardized protocol, then a "Flags for Review" list of anything not covered, ambiguous mappings and judgment calls. Do not omit any information from the original protocol.

Attributions

This guide was drafted and edited by Luciana Lozano and Raven Coffer, Cumulab, Boston University 2026 iGEM Team. Hosted by the DAMP Lab. Anthropic was used to help consolidate names of equipment.